

Algebra: Graphing Equations by Plotting Points

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Name: _____ Date: _____ Score: _____

DIRECTIONS

Complete the table of values for each equation, then identify the type of graph (line, parabola, etc.).

1 Find y when $x=3$:

$$y = 2x - 1$$

Answer: _____

2 Find y when $x=-2$:

$$y = 3x + 4$$

Answer: _____

3 Find y when $x=2$:

$$y = x^2 - 3$$

Answer: _____

4 Find y when $x=-1$:

$$y = x^2 + 2x$$

Answer: _____

5 Find y when $x=0$:

$$y = -2x + 5$$

Answer: _____

6 Find y when $x=4$:

$$y = (1/2)x - 1$$

Answer: _____

7 Find x-intercept:

$$y = 2x - 6$$

Answer: _____

8 Find y-intercept:

$$y = 3x - 9$$

Answer: _____

9 Find y when $x=-3$:

$$y = x^2 - x + 1$$

Answer: _____

10 Identify graph type:

$$y = x^2 + 2$$

Answer: _____



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Answer Key & Solutions

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TEACHER NOTES

Choose easy x-values (often -2, -1, 0, 1, 2). Substitute each into the equation and solve for y. Plot and connect the points.

ints.

1 $y=2x-1$, $x=3$

$= y = 5$

$y=2(3)-1=6-1=5$

2 $y=3x+4$, $x=-2$

$= y = -2$

$y=3(-2)+4=-6+4=-2$

3 $y=x^2-3$, $x=2$

$= y = 1$

$y=4-3=1$

4 $y=x^2+2x$, $x=-1$

$= y = -1$

$y=1-2=-1$

5 $y=-2x+5$, $x=0$

$= y = 5$

y-intercept: $y=5$

6 $y=1/2x-1$, $x=4$

$= y = 1$

$y=2-1=1$

7 $y=2x-6$ x-intercept

$= x = 3$

Set $y=0$: $0=2x-6$, $x=3$

8 $y=3x-9$ y-intercept

$= y = -9$

Set $x=0$: $y=-9$

9 $y=x^2-x+1$, $x=-3$

$= y = 13$

$y=9+3+1=13$

10 $y=x^2+2$

$= \text{Parabola}$

Highest power is 2 $\rightarrow$ parabola (opens up, vertex (0,2))



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